

Name:

BUID:

Workshop 5: Quadratic Forms

1. (*Review*) Determine the least-squares best-fit line for the data points $\{(0, 3), (1, 1), (-1, 1), (2, 3)\}$ using the least-squares method
2. (*Review*) Determine the design matrix and observation vector for finding the best-fit cubic polynomial for the data points in the previous problem
3. Determine an orthogonal diagonalization of $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$
4. Expand the quadratic form $Q(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ into a multivariate polynomial
5. Expand the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 0 \\ -1 & 0 & 5 \end{bmatrix} \mathbf{x}$ into a multivariate polynomial
6. Determine the symmetric matrix A such that $Q(x_1, x_2, x_3) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$ can be written as $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$
7. Determine if $Q(x_1, x_2, x_3) = 3x_1^2 + x_2^2 + 4x_2x_3 + x_3^2$ is positive definite, positive semidefinite, indefinite, or negative definite
8. Let $Q(\mathbf{x})$ be the quadratic form from problem 7. Determine $\max_{\|\mathbf{x}\|=1} Q(\mathbf{x})$ and $\min_{\|\mathbf{x}\|=1} Q(\mathbf{x})$
9. Let $Q(\mathbf{x})$ be the quadratic form from problem 7. Determine an orthogonal diagonalization of its underlying matrix
10. (*Optional challenge*) Give an example of a negative definite symmetric matrix in $\mathbb{R}^{2 \times 2}$ that has all positive entries (Hint. First determine its characteristic polynomial)

Write your solutions below and on the back on this page.

Continue your work here.